6 **(a)**	(coulomb is) ampere second	B1	[1]
(b)	(total) charge or $Q = nAle$	M1	
	$I = Q/t$ <u>and</u> $l/t = v$	M1	
	$I = nAle/t = nAve$ therefore $v = I/nAe$	A1	[3]
(c) **(i)**	ratio $= (I/nA_Ye)/(I/nA_Ze)$	C1	
	$= A_Z/A_Y$ or $4A/A$ or $\pi d^2/(\pi d^2/4)$	C1	
	$= 4$	A1	[3]
(ii)	$R = \rho l/A$ or $R = 4\rho l/\pi d^2$	B1	
	$R_Y = \rho l/A$ <u>and</u> $R_Z = \rho(2l)/4A$ so $R_Y/R_Z = 2$ *or* $R_Y = 4\rho l/\pi d^2$ <u>and</u> $R_Z = 4\rho(2l)/\pi 4d^2$ or $2\rho l/\pi d^2$ so $R_Y/R_Z = 2$	A1	[2]
(iii)	$V = 12R_Y/(R_Y + R_Z)$ or $I = 12/(R_Y + R_Z)$ <u>and</u> $V = IR_Y$	C1	
	$V = 12 \times 2/3$		
	$= 8(.0)\,\text{V}$	A1	[2]
(iv)	ratio $= I^2R_Y/I^2R_Z$ or $(V_Y^2/R_Y)/(V_Z^2/R_Z)$ or $(V_YI)/(V_ZI)$		
	$= 2$	A1	[1]